

Understanding Key Success Factors in Aerosol Microphysics Emulation

BER-ASCR SciDAC Partnership (PAESCAL); SciDAC Institute (LEADS)

Scientific Achievement

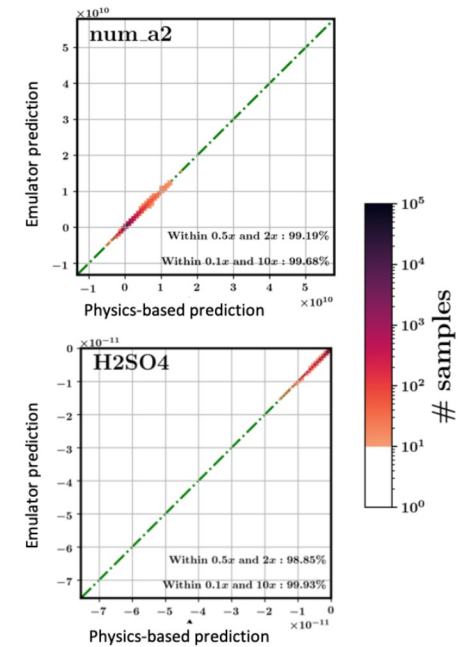
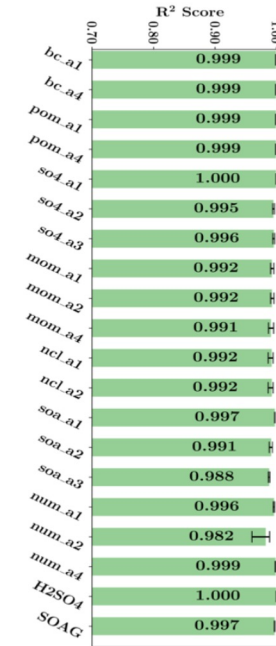
- Identified three key factors controlling high-accuracy ML emulation of aerosol microphysics in Energy Exascale Earth System Model (E3SM) version 2: training convergence, network size, and variable transformation.

Significance and Impact

- Despite strong interests in using ML to emulate the aerosol life cycle, there is limited understanding of how emulator design choices affect performance.
- This work demonstrates that, with proper scaling and convergence, a moderately-sized simple network can predict aerosol concentration changes with high accuracy, showing promise of avoiding higher computational costs needed by more complex ML approaches.
- Results from this work provide practical guidance for building accurate and efficient emulators for aerosol microphysics, with insights likely transferable to other multi-scale problems.

Technical Approach

- Construct ML emulators for cloud-free aerosol microphysics in the Modal Aerosol Module (MAM) in E3SM, using a feedforward neural network to study fundamental design choices.
- Systematically evaluate key emulator design factors—including training convergence behavior, network complexity, and variable transformation—and quantify their impact on emulation accuracy.



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Collaborating Institutions: PNNL, LBNL

ASCR Program: SciDAC (partnerships, Institutes)

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